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Medical

Diabetes Data Analysis

We're helping patients control diabetes and prevent complications through care. Our diabetes management program provides comprehensive support for better health.



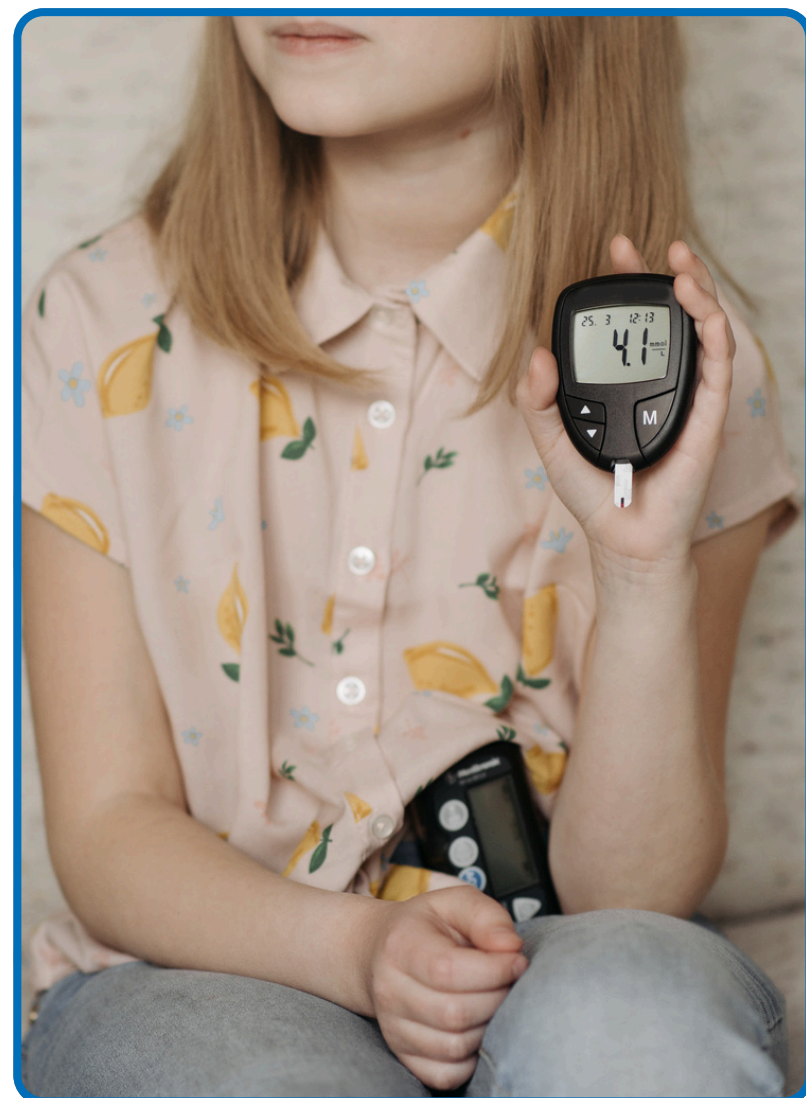
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More Than
50+
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Diabetes Data Analysis



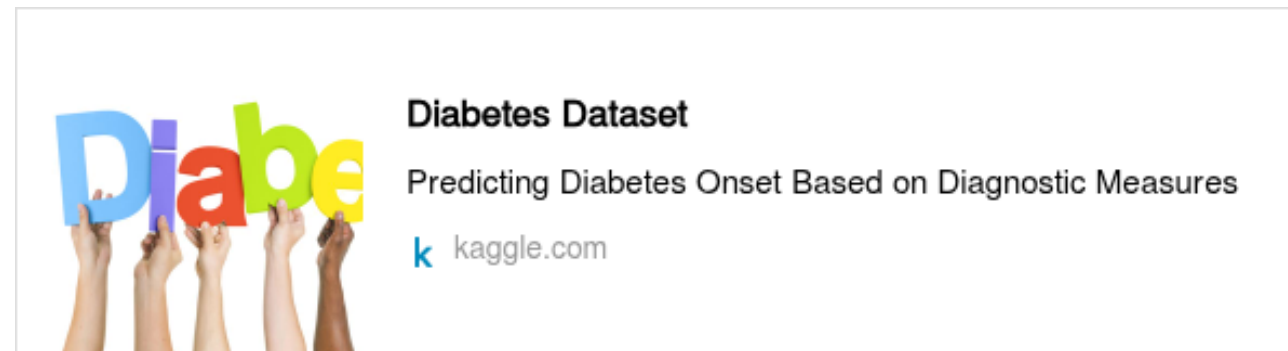
An end-to-end data analysis project that explores diabetes-related patterns, risk factors, and patient outcomes using multiple data analysis tools.

- **Analyze diabetes patient data**
- **Identify important risk factors**
- **Build interactive dashboards**
- **Support data-driven healthcare insights**





Dataset Overview & Source



Source: Kaggle Platform

- **Dataset Size: 768 patient records**
- **The dataset contains 768 rows and 9 columns.**
- **Key health indicators include Glucose, BMI, BloodPressure, and Age**
- **The target variable is Outcome, representing diabetes presence.**





Methodology & Data Processing

- **Data Preprocessing (Python):**

Handled missing values, verified data types, and managed feature distributions.

- **Data Analysis (SQL):**

Executed aggregations and structured queries to extract core health metrics.

- **Exploratory Analysis & Visualization:**

Evaluated statistical variances, correlations, and prepared data for dashboards.

- **Tech Stack:**

Python, SQL, and Visualization Tools.





Key Results & Statistical Insights

Population Overview:

Total of 768 patient records analyzed, with a confirmed diabetes rate of 34.90% (268 diabetic cases).

Key Health Averages (Diabetic vs. Non-Diabetic):

Glucose: Significantly higher in diabetic patients (142.30 mg/dL) compared to non-diabetic (110.62 mg/dL).

BMI: Higher average among diabetic individuals (35.26) vs. healthy ones (30.83).

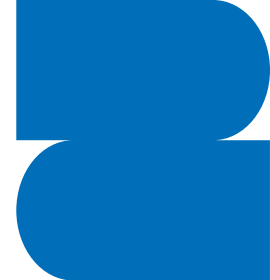
High-Risk Intersections:

Patients with High Glucose (>125) & Obesity ($\text{BMI} \geq 30$) show a severe diabetes risk rate reaching 70.32%.

Age & Demographic Impact:

The 45–54 age group records the highest diabetes prevalence across the population at 60.76%.





Visual Analytics & Interactive Dashboards

Transition to Visualization:

Bridging analytical insights with interactive visualization to monitor diabetes patterns and clinical metrics dynamically.

Integrated Tech Stack:

Leveraging Microsoft Excel for foundational descriptive layouts and Power BI for advanced multidimensional business intelligence and filtering.



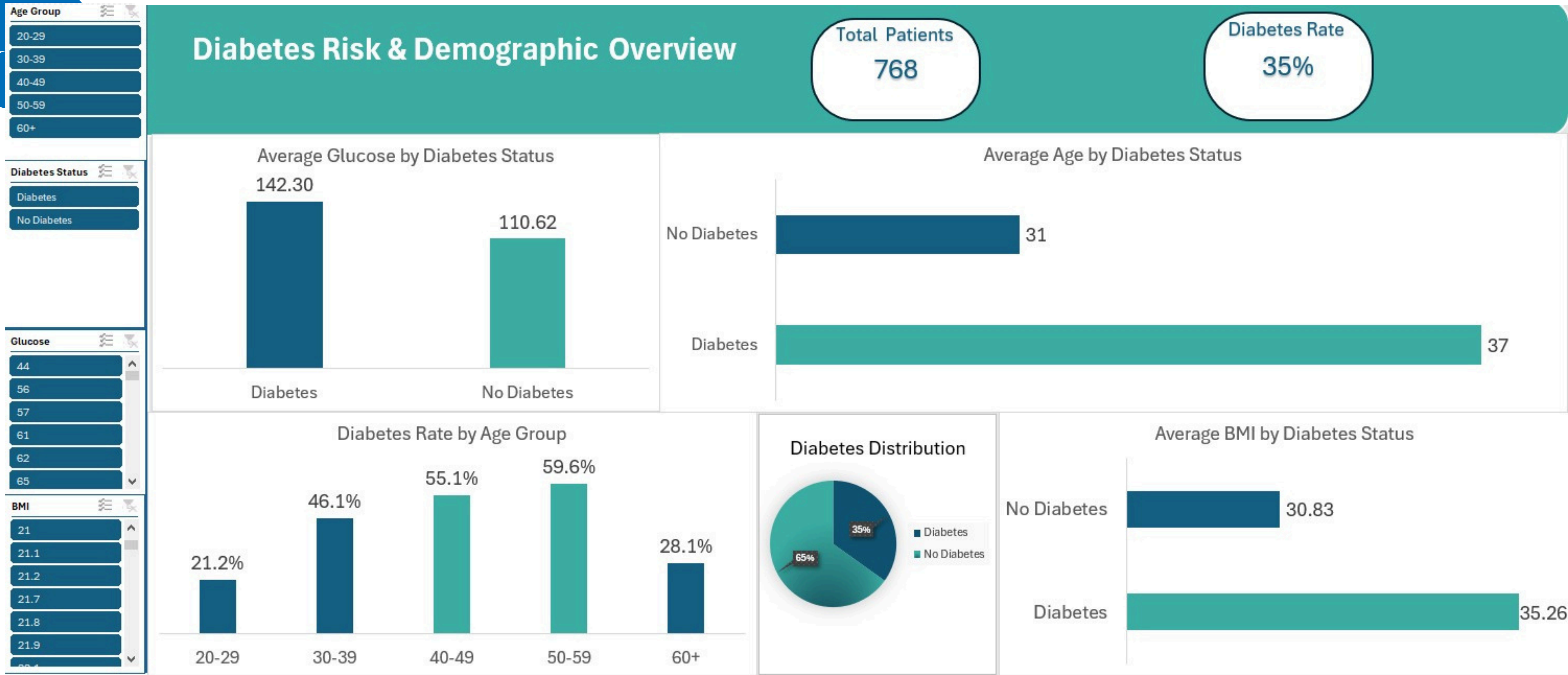


Diabetes Health Analytics

Better Insights • Healthier Lives

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- Comprehensive Project Goal:

To analyze diabetes risk factors by connecting health metrics (Glucose & BMI) with patient age to guide clinical and preventive decisions.
- Target Audience & Stakeholders:

Medical Teams: To evaluate patient health metrics and individual risk levels.

Healthcare Analysts: To track underlying patterns and disease prevalence across demographics.

Hospital Management: To plan clinical resources and target preventive health programs efficiently

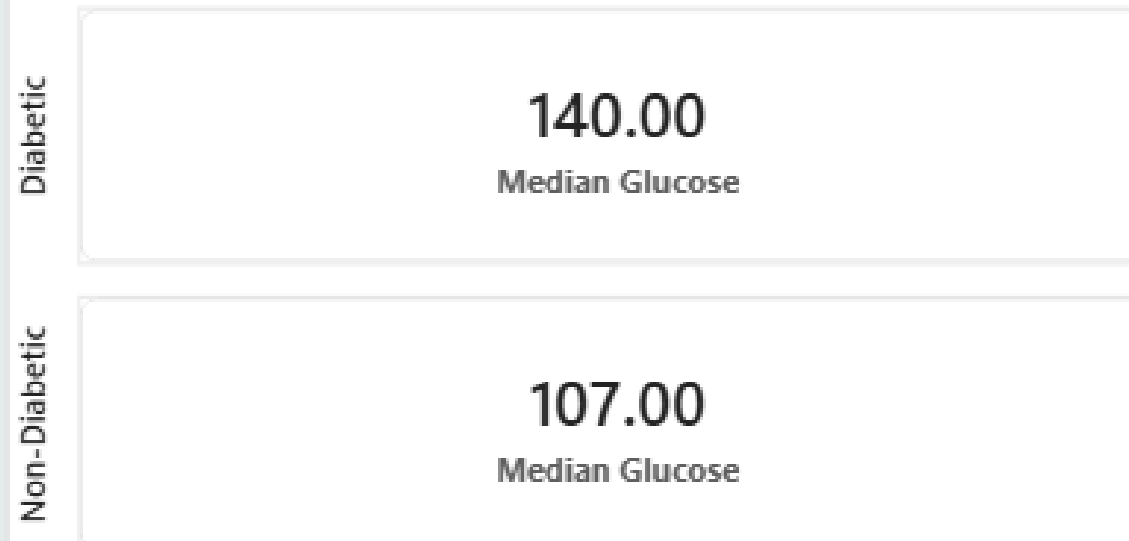


Glucose & Risk Factor Analysis

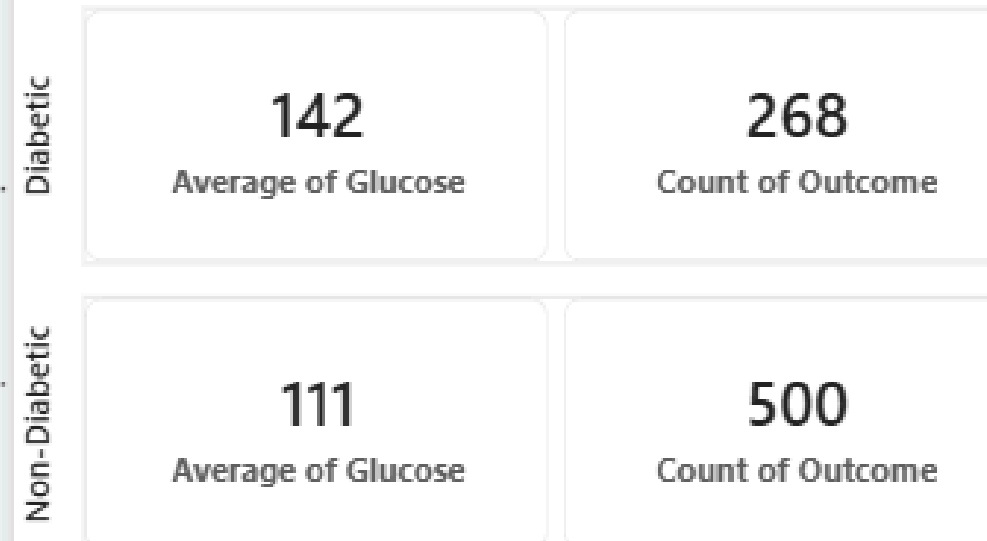
Diabetes Rate (High Glucose)
59.53%

Glucose & Outcome Correlation Coefficient
0.496

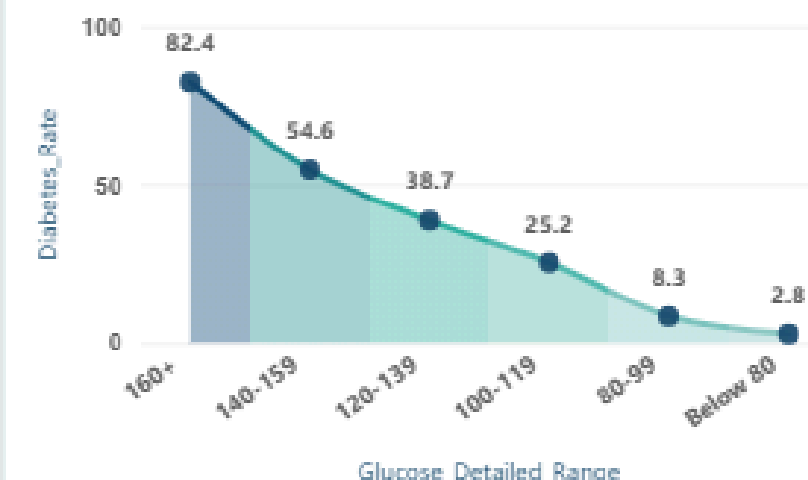
Median Glucose Level by Diagnosis Group



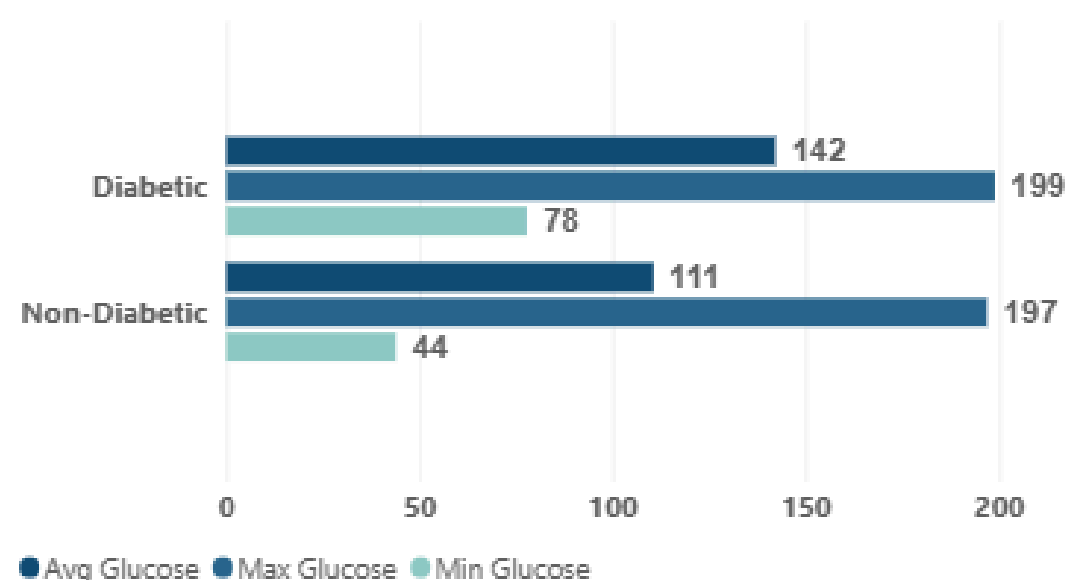
Average Glucose Level by Diagnosis Group



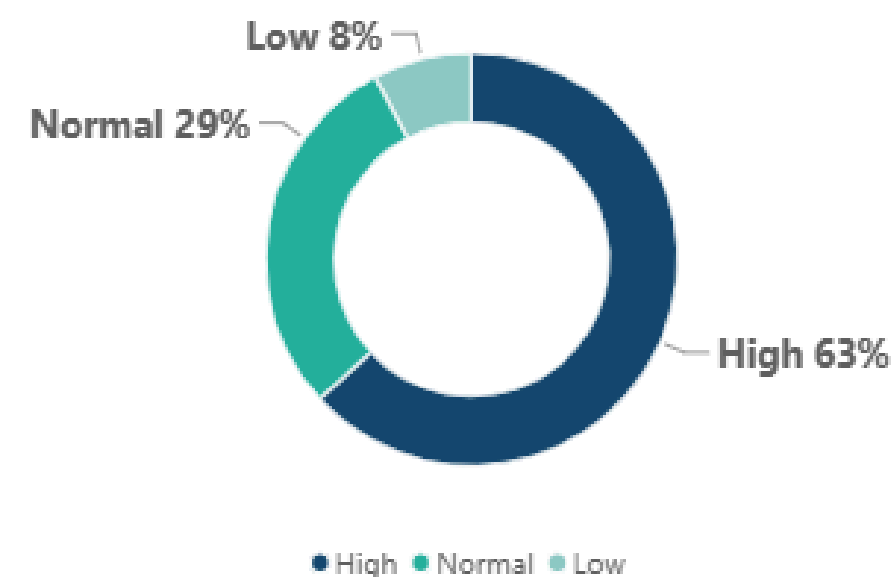
Diabetes Rate by Glucose Range



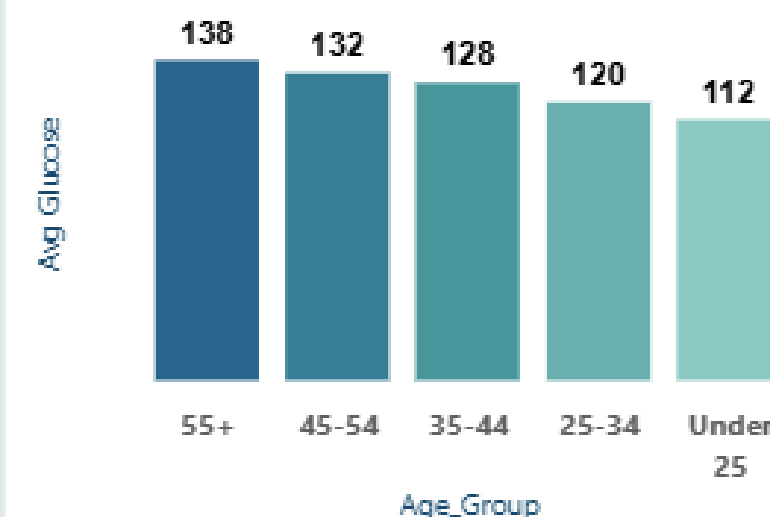
Avg, Max and Min Glucose by Outcome



Patient Distribution by Glucose Risk Category



Average Glucose Level Across Age Groups



Glucose_Range

All

Diabetes_Status

All

Welcome

Clinical Vitals

Risk Matrix

Glucose & Risk

Target Audience&Goal

Comprehensive Dashboard Goal(Previous):

To analyze the direct statistical and correlational relationship between patient glucose levels and diabetes diagnosis by evaluating metrics such as minimum/maximum limits, averages, medians, detailed glycemic risk ranges, and correlation coefficients to identify critical blood sugar thresholds for clinical assessment.

Comprehensive Dashbord 3 Goal(Next):

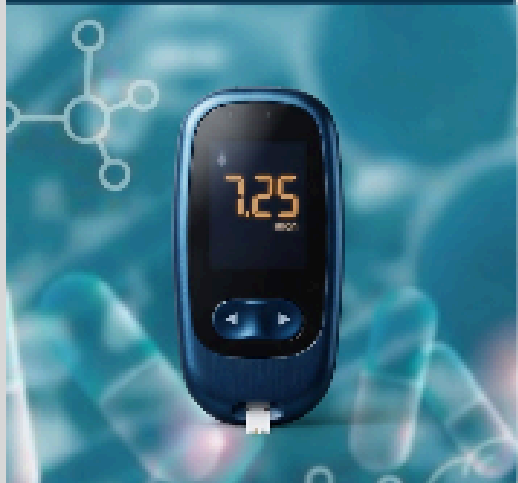
To analyze the physiological risk factors of diabetes by evaluating physical and metabolic metrics—specifically focusing on BMI categories, blood pressure distributions, insulin levels, and their combined impact on disease prevalence to optimize clinical risk assessment and preventative care.

Target Audience & Stakeholders:

Medical Teams: To evaluate patient glycemic profiles, compare glucose baselines and distribution ranges between diabetic and non-diabetic groups, and assess individual metabolic risk.

Healthcare Analysts: To track underlying glycemic patterns, calculate statistical correlation coefficients, and analyze disease prevalence across segmented glucose thresholds.

Hospital Management: To plan clinical screening protocols, allocate endocrinology resources efficiently, and target preventative health programs toward high-risk glycemic segments



BMI, Insulin & Clinical Indicators

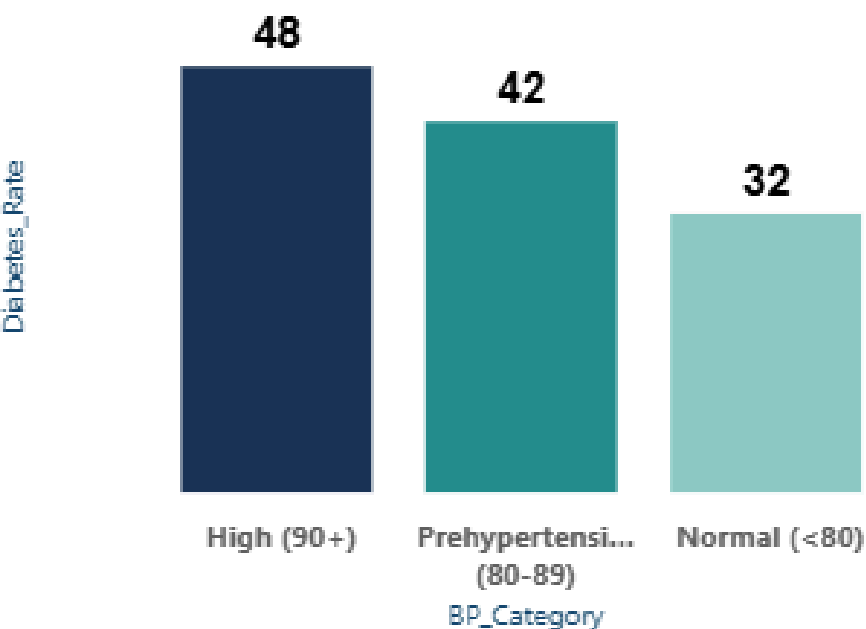
Diabetes Rate
35

Total Diabetic Patients
268

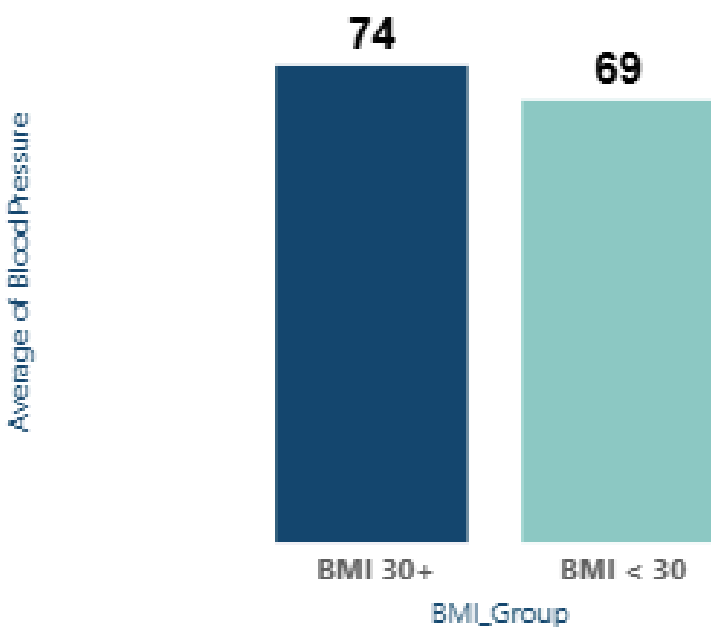
Total Patient Sample
768

High Risk Diabetes Rate
62.30

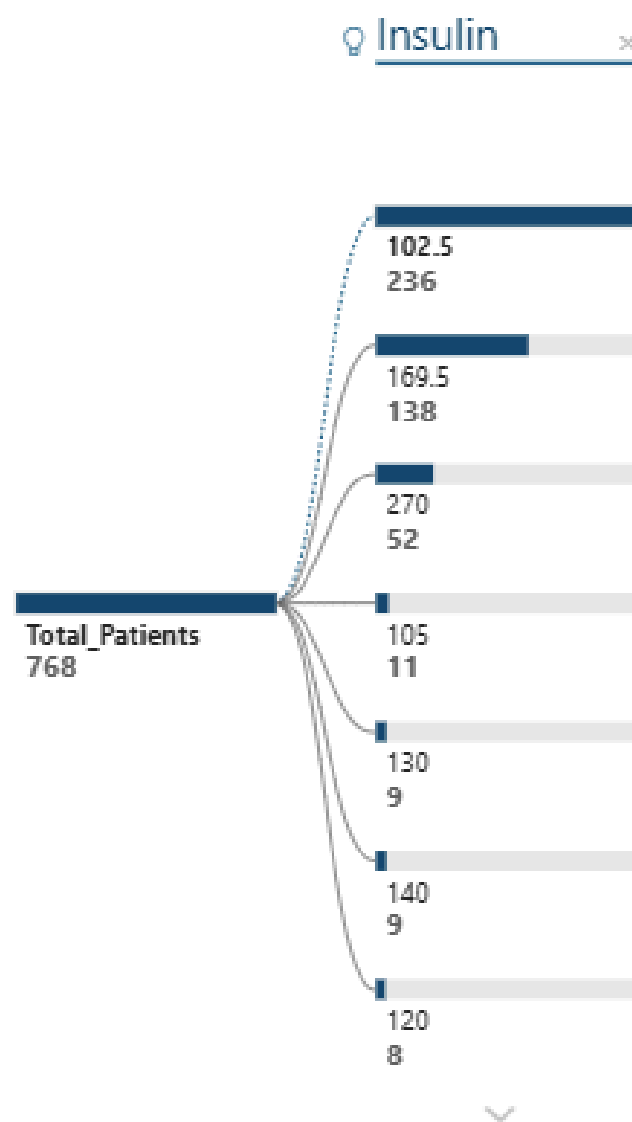
Diabetes Prevalence by Blood Pressure Category



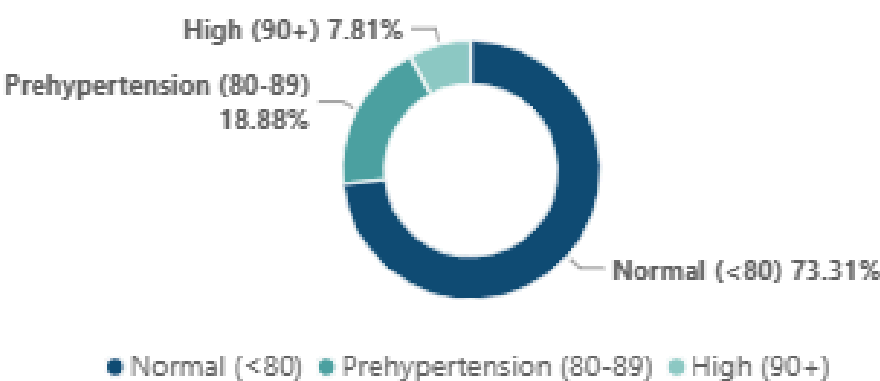
Average Blood Pressure Across BMI Groups



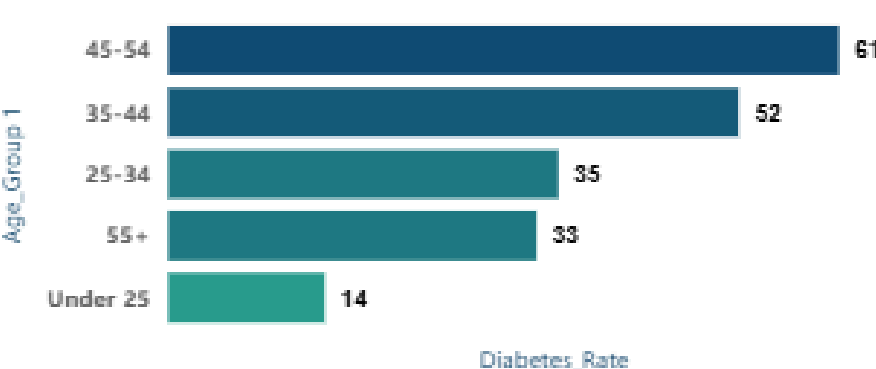
Patient Distribution & Risk by Insulin Levels



Blood Pressure Distribution Among Patients



Diabetes Rate Across Age Groups



BP_Category
All

BMI_Group
All

Welcome

Clinical Vitals

Risk Matrix

Glucose & Risk



Age_Group 1

All

Diabetes_Status

All

Welcome

Glucose & Risk

Clinical Vitals

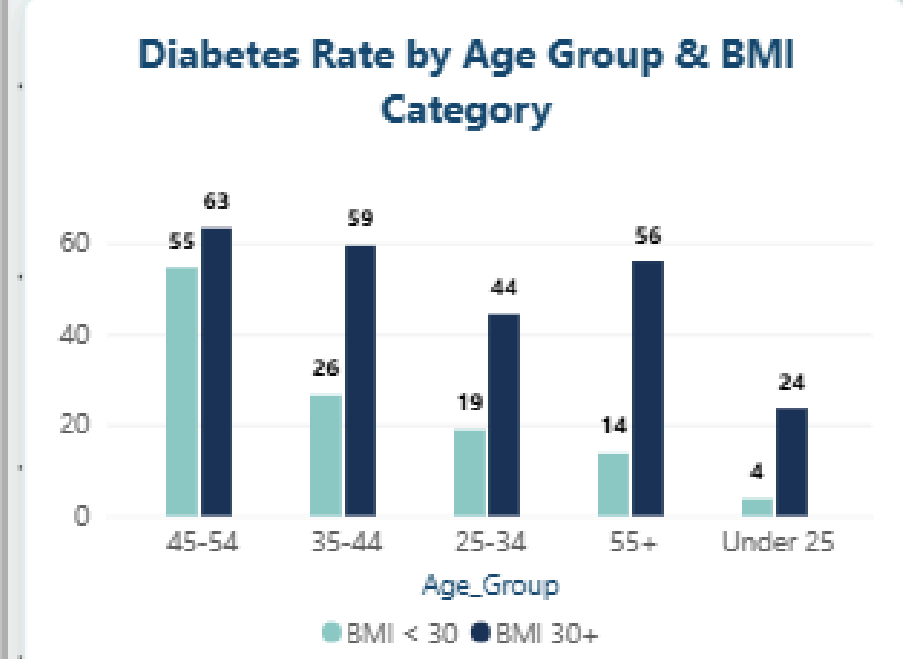
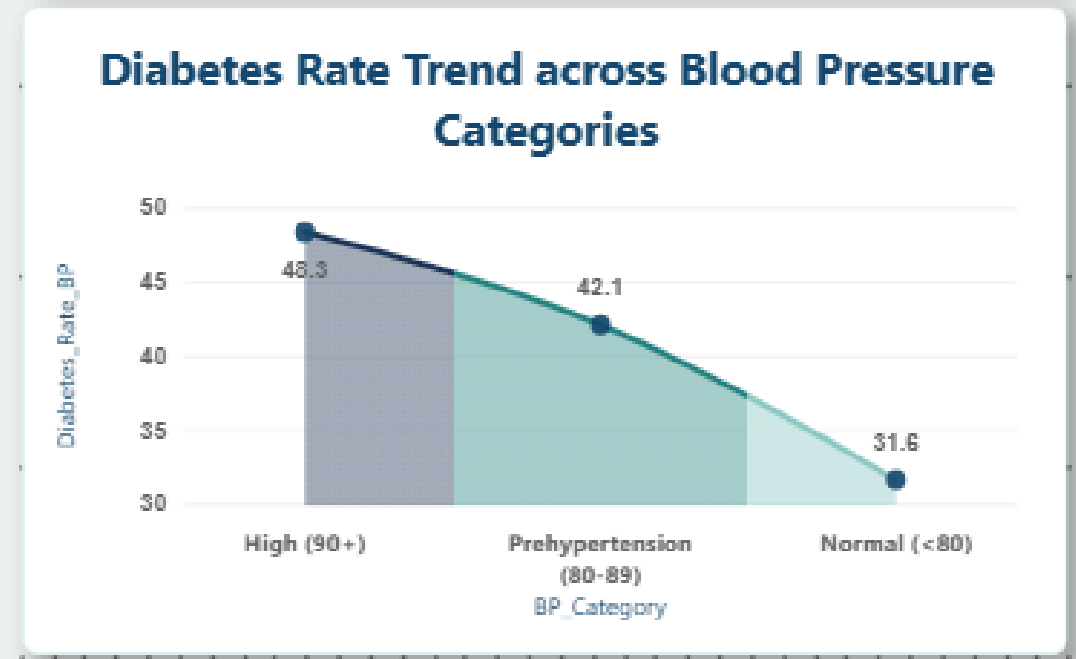
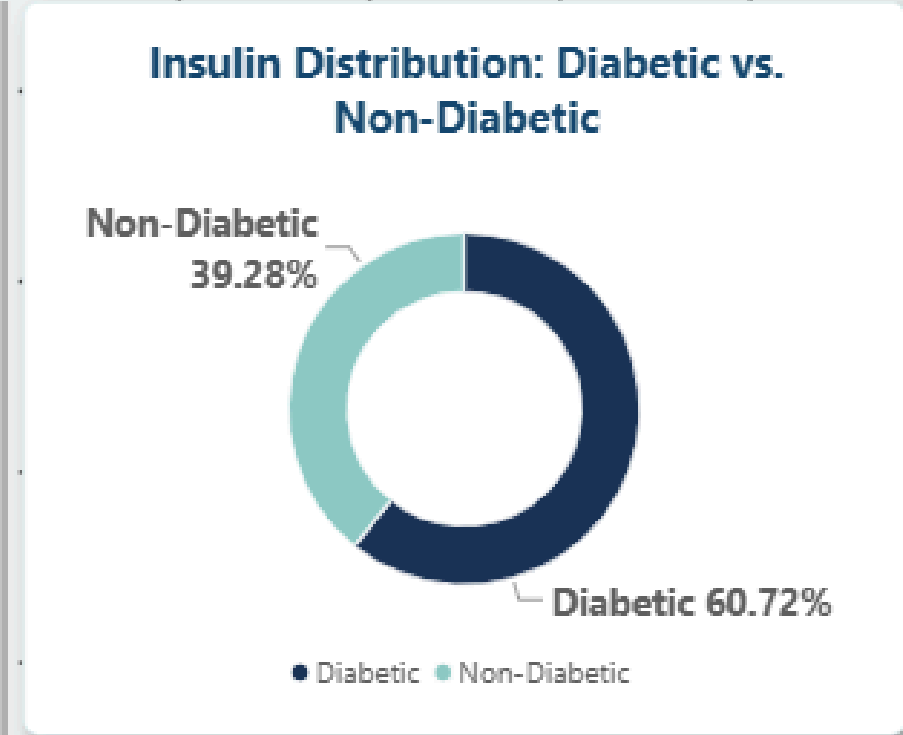
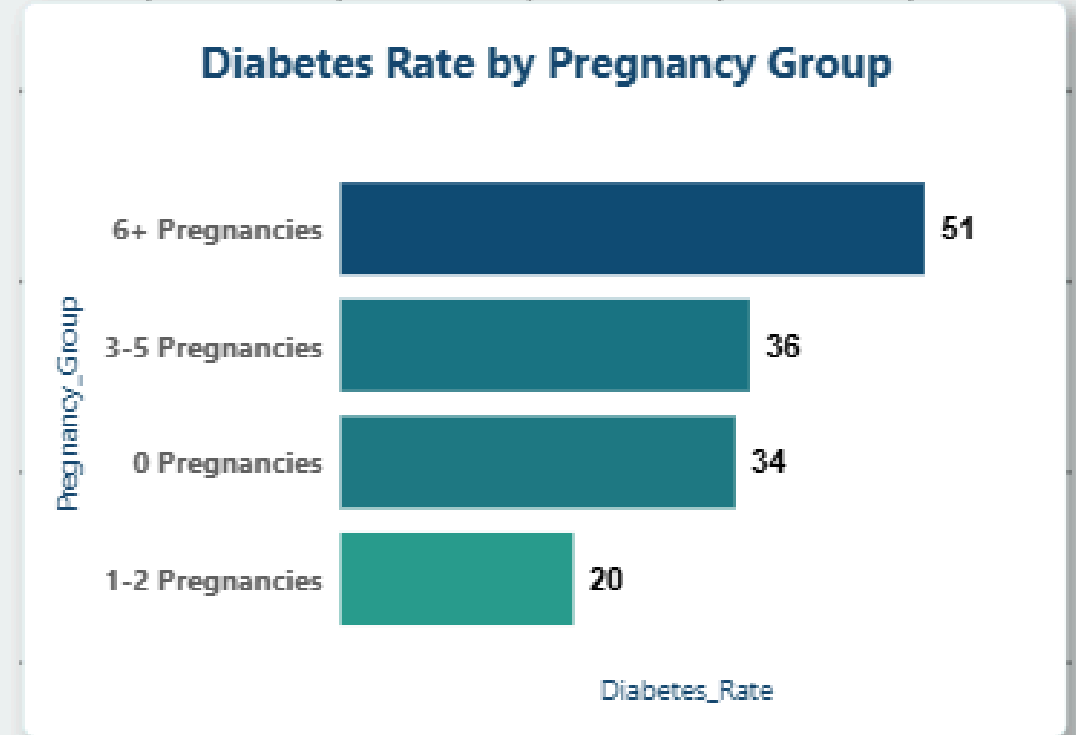
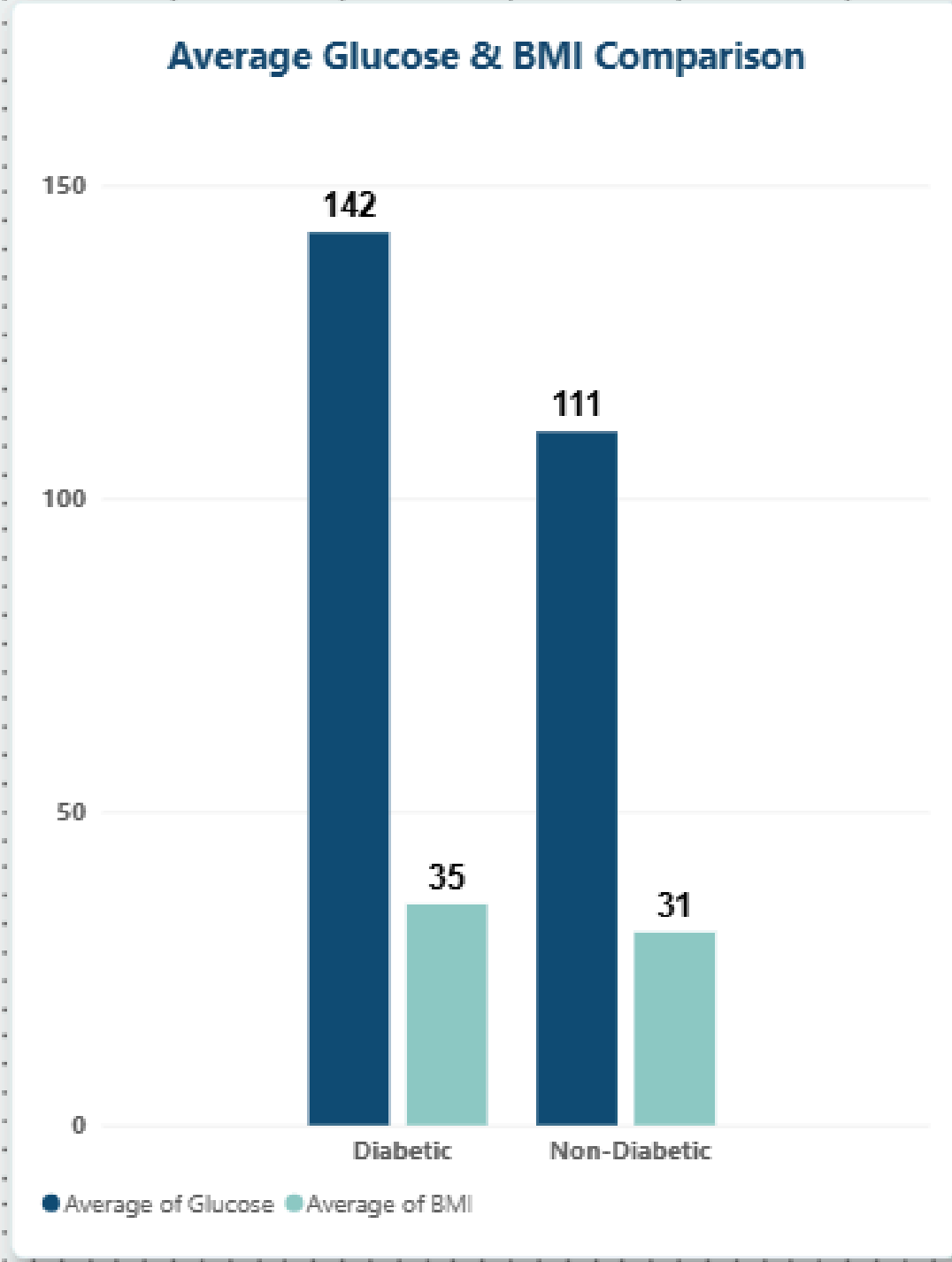
Multi-Factor Risk Analysis

Diabetes Rate
35

Total Diabetic Patients
268

Total Patient Sample
768

High Risk Diabetes Rate
62.30



Target Audience&Goal

Comprehensive Dash bord 4 Goal(previos):

To evaluate compound health risks by cross-analyzing multiple clinical variables simultaneously—specifically measuring how combinations of elevated glucose, high BMI, pregnancy frequencies, and insulin levels amplify diabetes rates—to uncover complex risk patterns and patient demographics

Target Audience & Stakeholders:

Medical Teams: To evaluate patient glycemic profiles, compare glucose baselines and distribution ranges between diabetic and non-diabetic groups, and assess individual metabolic risk.

Healthcare Analysts: To track underlying glycemic patterns, calculate statistical correlation coefficients, and analyze disease prevalence across segmented glucose thresholds.

Hospital Management: To plan clinical screening protocols, allocate endocrinology resources efficiently, and target preventative health programs toward high-risk glycemic segments



Final Insights & Key Takeaways

Primary Risk Driver:

Clinical indicators demonstrate that the simultaneous presence of elevated blood glucose and obesity (BMI ≥ 30) serves as the core catalyst for diabetes onset.

Vulnerable Demographic:

Epidemiological distribution confirms that middle-aged adults, specifically the 21-40 age bracket, exhibit the highest susceptibility and disease prevalence.

Analytical Value:

The integration of automated preprocessing workflows and complex SQL queries successfully bridged raw patient records with actionable clinical intelligence.

Future Outlook:

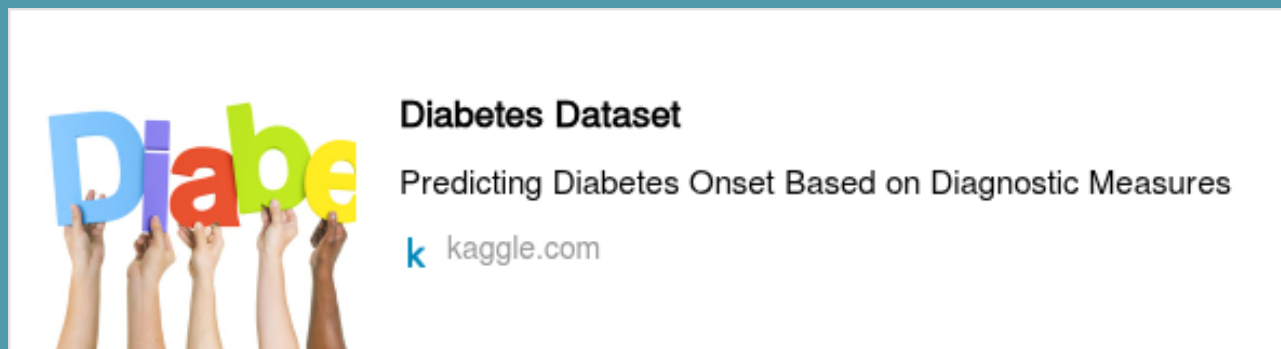
These validated insights establish a robust baseline architecture for deploying advanced predictive Machine Learning models in subsequent phases.





Control Your Diabetes Today

Thank You



Faculty of Computers and Artificial Intelligence

